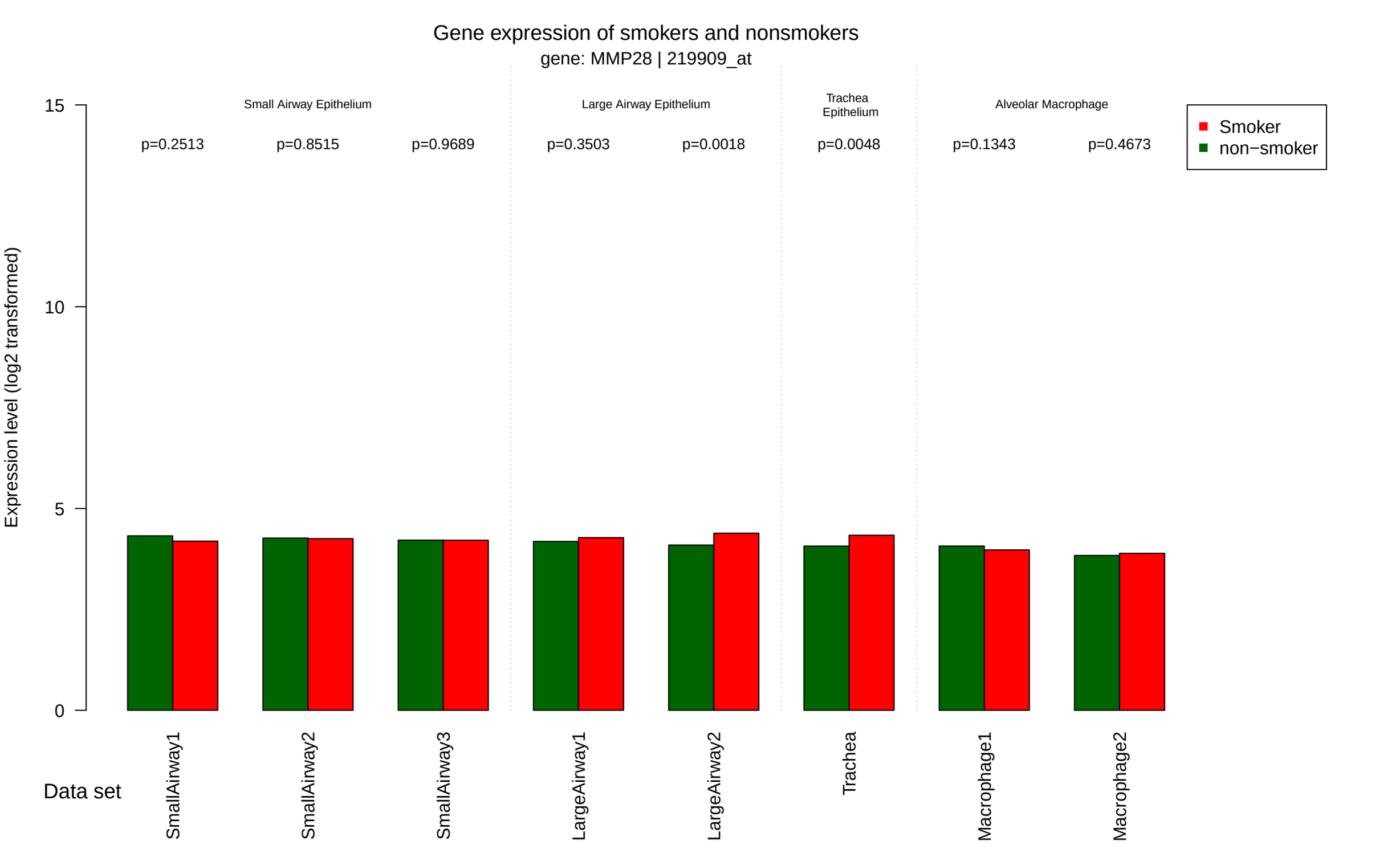
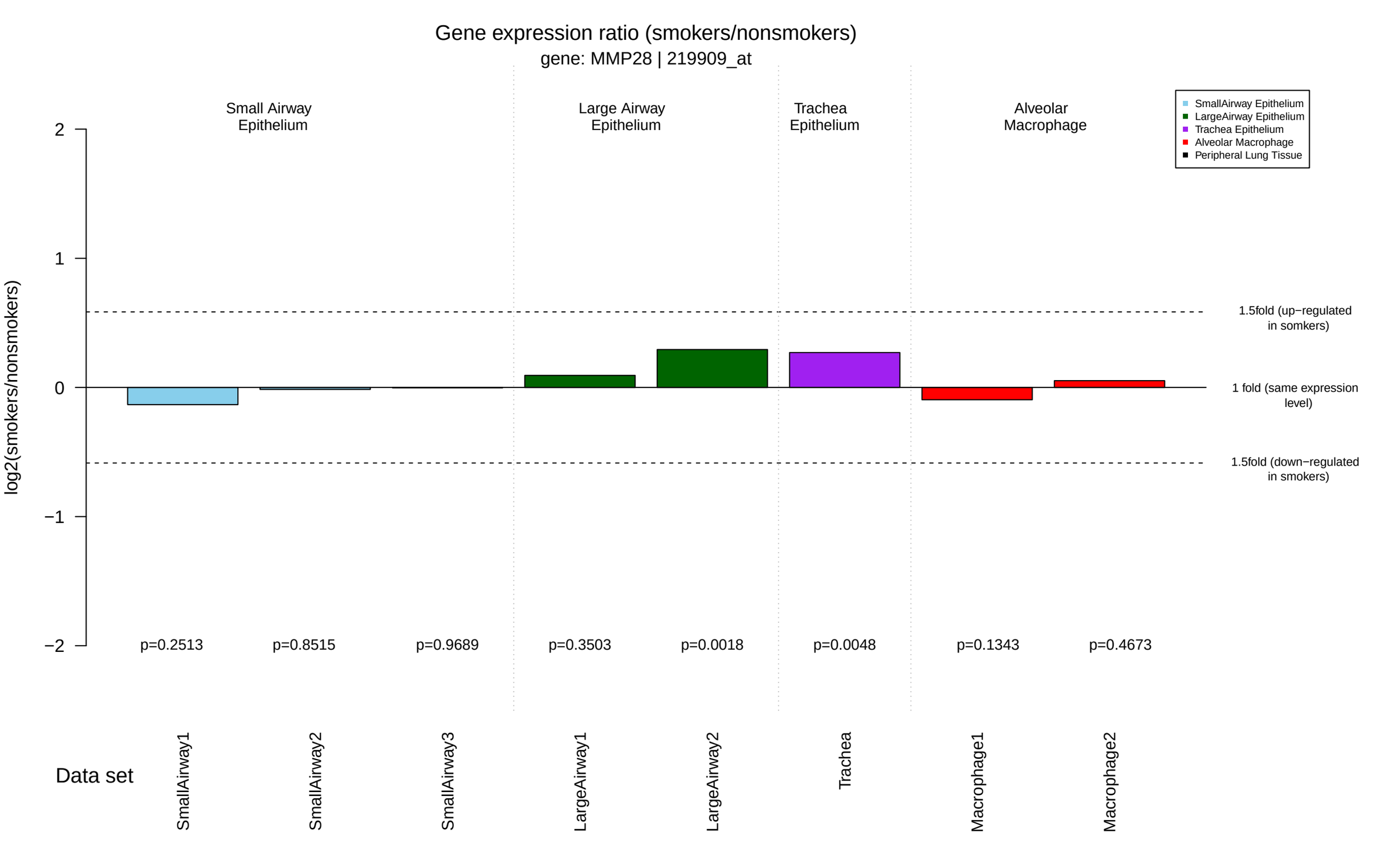
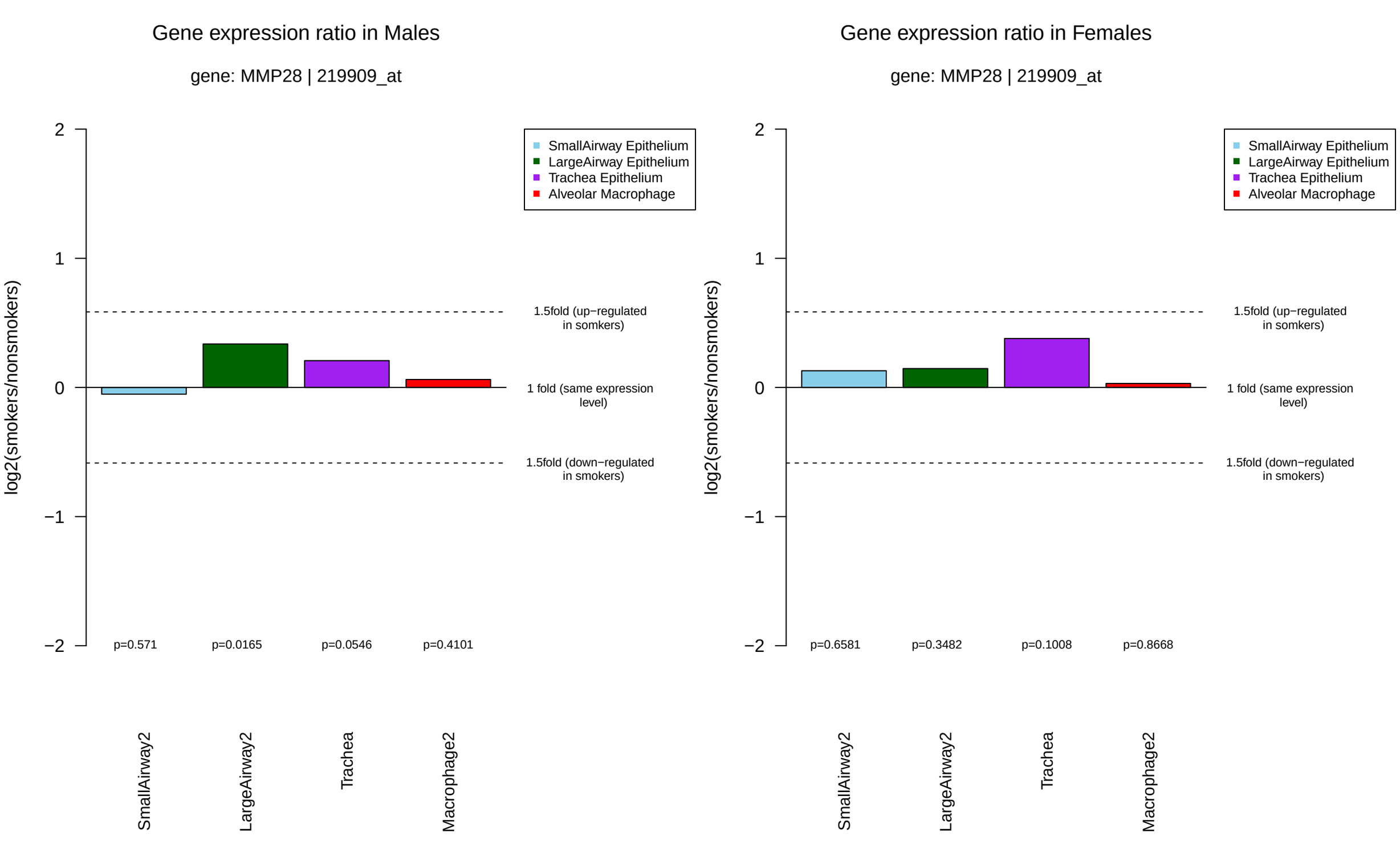
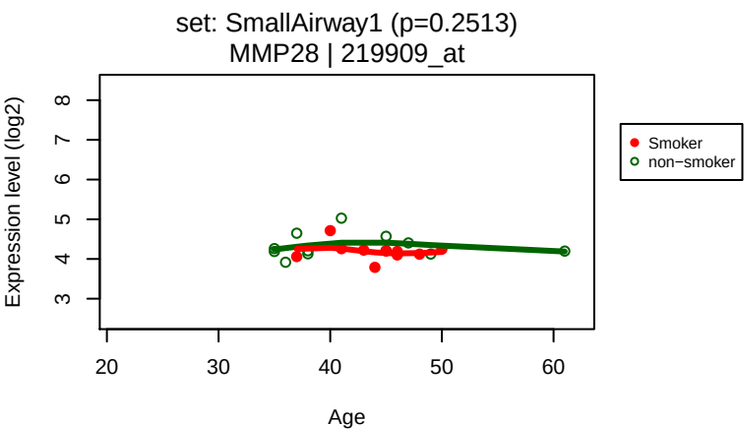
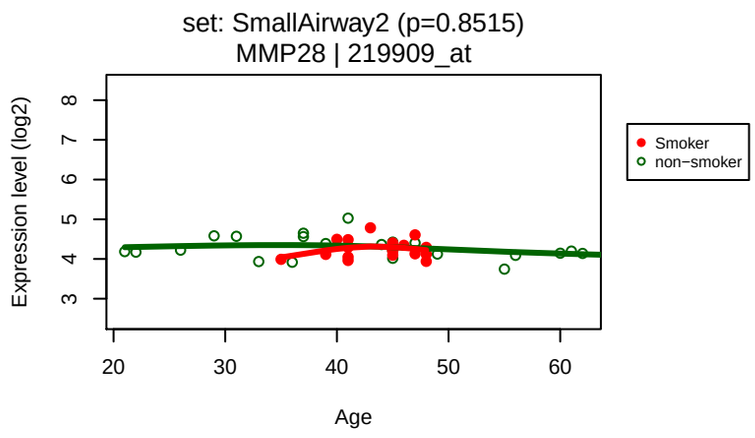
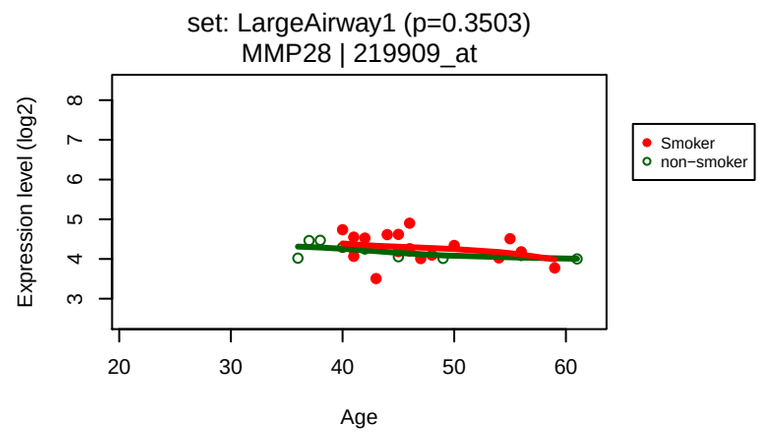

Gene expression of smokers and nonsmokers
gene: MMP28 | 219909_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2513
p=0.8515
p=0.9689
p=0.3503
p=0.0018
p=0.0048
p=0.1343
p=0.4673
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

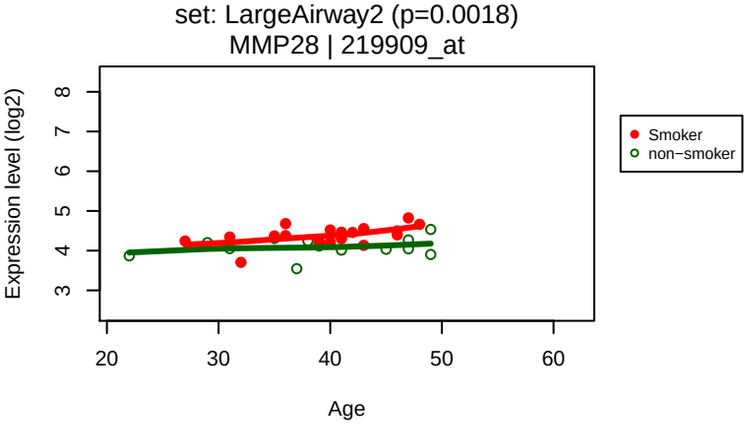

Gene expression ratio (smokers/nonsmokers)
gene: MMP28 | 219909_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2513
p=0.8515
p=0.9689
p=0.3503
p=0.0018
p=0.0048
p=0.1343
p=0.4673
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

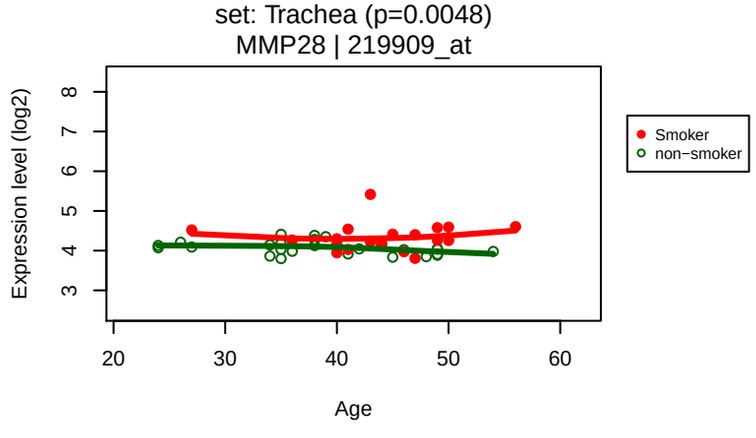

Gene expression ratio in Males
gene: MMP28 | 219909_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.571
p=0.0165
p=0.0546
p=0.4101
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP28 | 219909_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6581
p=0.3482
p=0.1008
p=0.8668
SmallAirway2
LargeAirway2
Trachea
Macrophage2

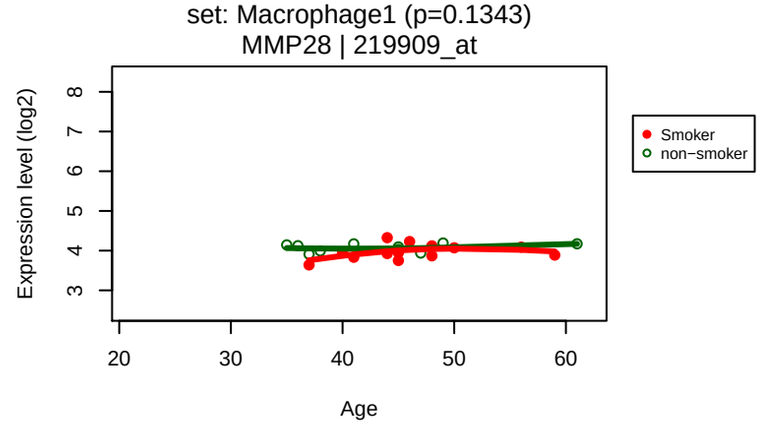

set: SmallAirway1 (p=0.2513)
MMP28 | 219909_at
Expression level (log2)
Age
Smoker
non−smoker

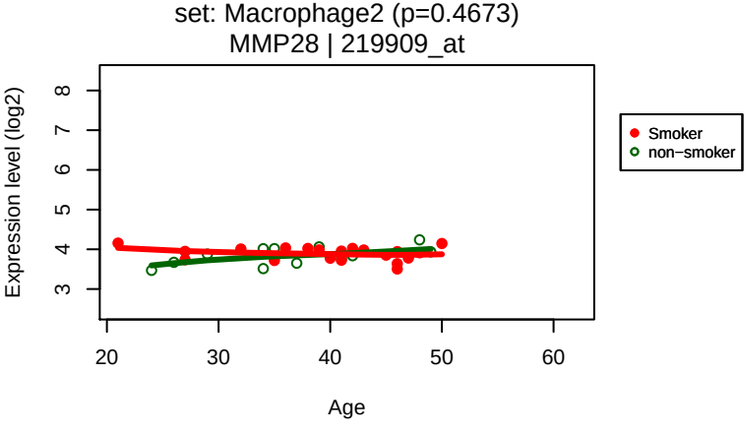

set: SmallAirway2 (p=0.8515)
MMP28 | 219909_at
Expression level (log2)
Age
Smoker
non−smoker

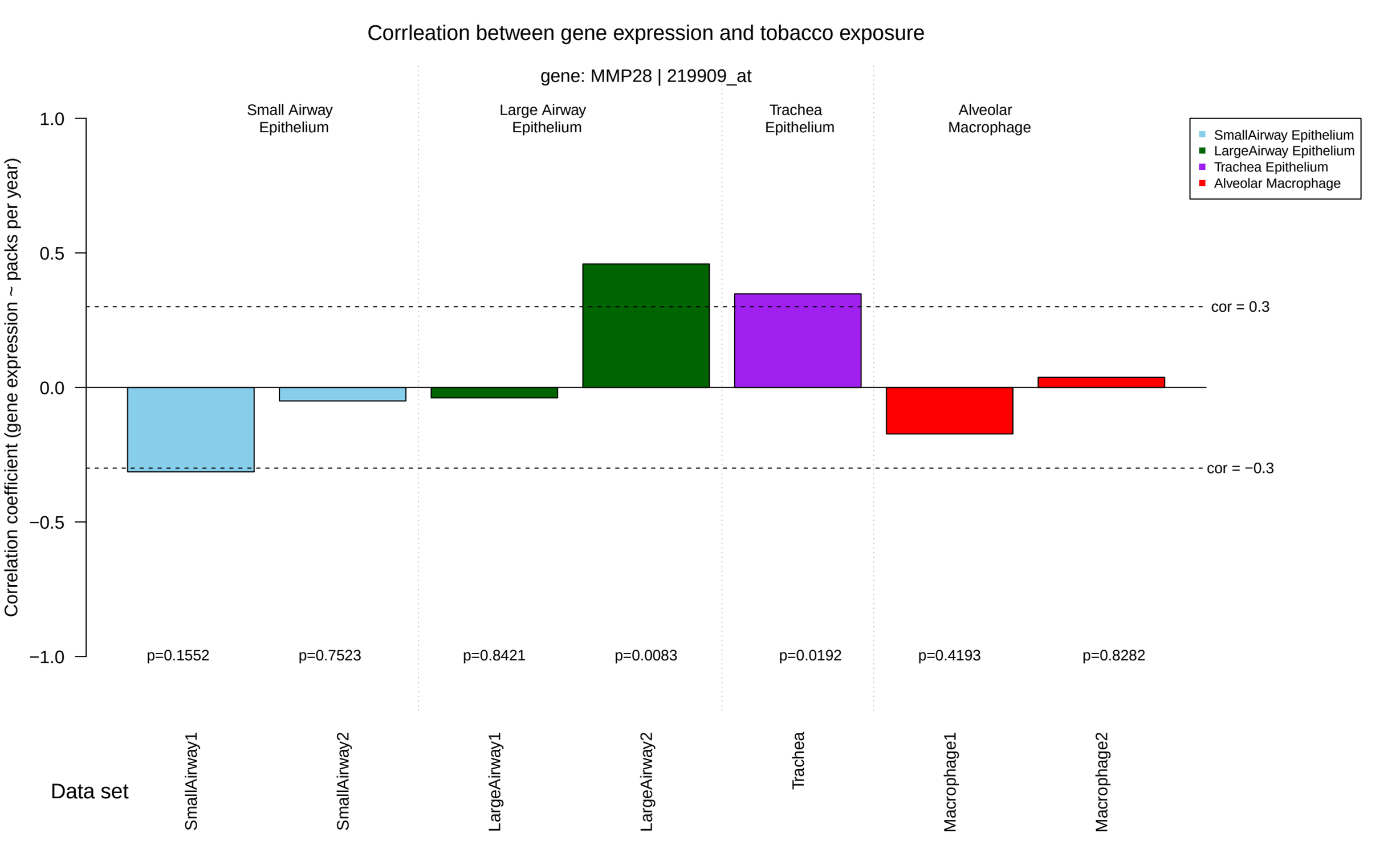

set: LargeAirway1 (p=0.3503)
MMP28 | 219909_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0018)
MMP28 | 219909_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0048)
MMP28 | 219909_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1343)
MMP28 | 219909_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4673)
MMP28 | 219909_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP28 | 219909_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1552
p=0.7523
p=0.8421
p=0.0083
p=0.0192
p=0.4193
p=0.8282
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2